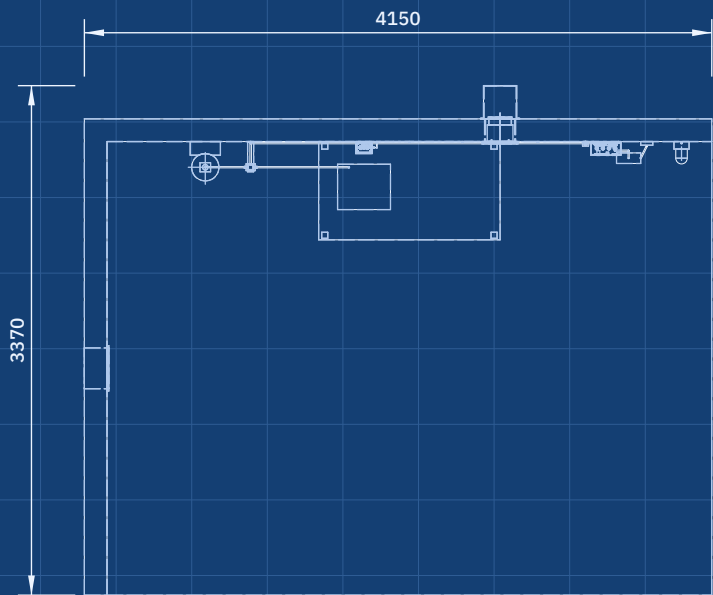


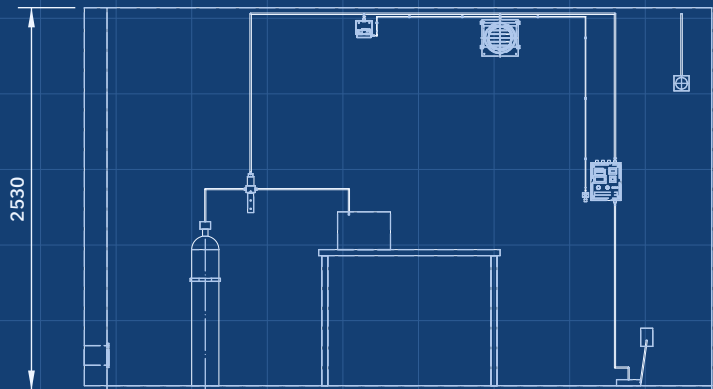
CONCEPT, NOT FOR FABRICATION

H2Guard · [github.com/BoujeeEnjinia1701/h2guard](https://github.com/BoujeeEnjinia1701/h2guard)



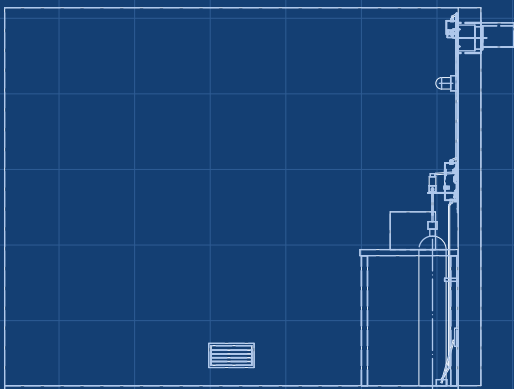
TOP VIEW

Scale 1:50; looking down (along -Z)



FRONT VIEW

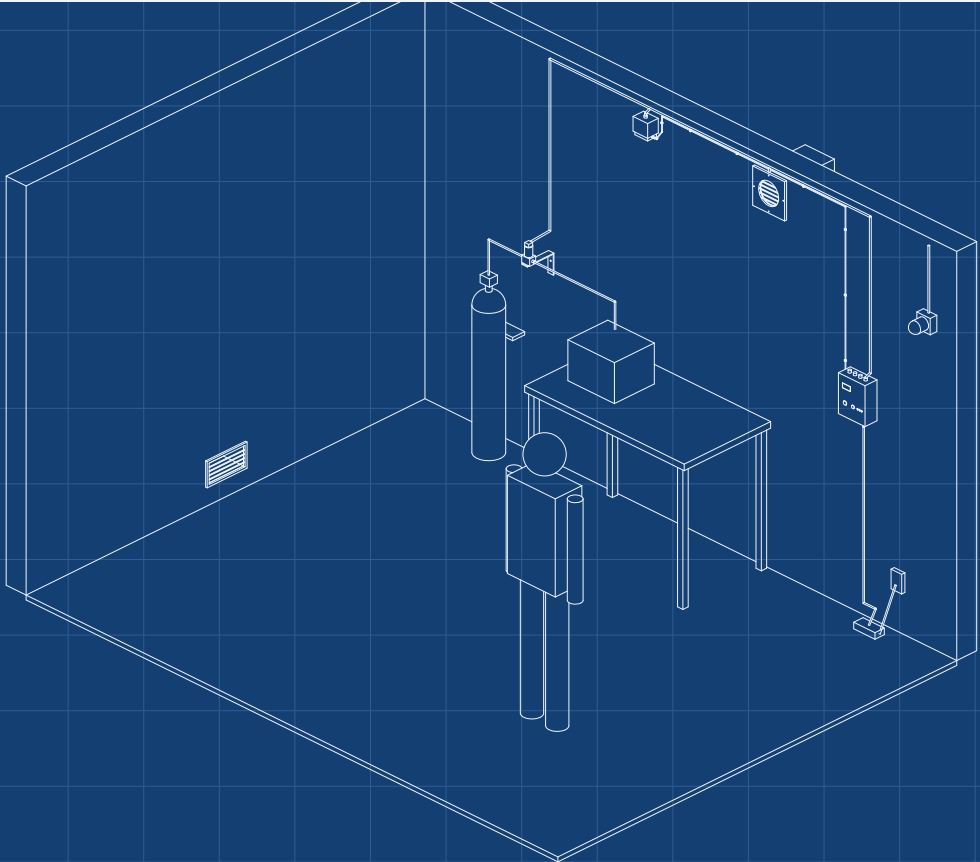
Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)

| REV | DESCRIPTION                     | DATE       | BY |
|-----|---------------------------------|------------|----|
| P1  | Concept sheet                   | 2026-09-25 | AC |
| P2  | From the TRL 3 parametric model | 2026-09-25 | AC |
| P3  | Key figures after DDR-002       | 2026-09-25 | AC |
| P4  | Constructable design (DDR-003)  | 2026-10-01 | AC |



ISOMETRIC VIEW

Not to scale; figure is a 1.75 m person; seen from the front right and above, about 30 deg elevation

KEY FIGURES

Warn at 10 % LFL (0.4 % vol H<sub>2</sub>); trip at 25 % LFL (1.0 % vol)  
Leak to valve closed 35.7 s with t<sub>90</sub> 30 s assumed (HGD-CAL-001)  
Exhaust 150 m<sup>3</sup>/h continuous, 347 m<sup>3</sup>/h boost; room 5 % LFL  
Fail-safe: NC valve, series hardware trip relay, 1 s watchdog  
24 V DC build; mains only inside a certified supply  
Warning held 5 min also closes the valve (DDR-002)  
Estimated cost USD 289; value-engineering target USD 265

H2GUARD

[github.com/BoujeeEnjinia1701/h2guard](https://github.com/BoujeeEnjinia1701/h2guard)

Leak detector and interlock concept

|                                   |                    |                                                                                                     |                           |                              |
|-----------------------------------|--------------------|-----------------------------------------------------------------------------------------------------|---------------------------|------------------------------|
| DRAWING NO.<br><b>HGD-DWG-010</b> |                    | REV<br><b>P4</b>                                                                                    | SHEET<br><b>1 of 1</b>    | SIZE<br><b>ANSI B</b>        |
| SCALE<br><b>1:50</b>              | UNITS<br><b>mm</b> | PROJECTION<br> | DATE<br><b>2026-10-01</b> | DRAWN<br><b>Amish Chadha</b> |

MATERIAL / NOTES  
Built from cad/src/model.py; room, bench and gas store are context. GA is HGD-DWG-001